

Attachment # 2 – Integrated Visual Augmentation System (IVAS) Next Request for Information - Contents

Vendors should submit an RFI response describing their IVAS Next System solution, as outlined below.

Executive Summary (EXSUM). Vendors should provide a one (1) page summary of their IVAS Next technical approach, key features and expected performance, projected benefits, and technology readiness level.

1.0. Technical Approach. Vendors should submit a narrative response which describes the following:

1.1. System Design & Features. The design and features for their IVAS Next solution. The narrative should cover Heads Up Display (HUD), body-worn compute, night vision capability, company-level communications network, and any software supporting services or systems (e.g., Cloud or Edge computing).

1.1.1. Human-Machine Integration (HMI). How their design approach enables HMI.

1.1.2. Human Factors. How their design incorporates human factors.

1.1.2.1 Visual Comfort. How their design mitigates known visual comfort limitations, to include avoidance of visual discomfort and nausea from virtual, mixed, and augmented reality headsets.

1.1.3. Situational Awareness Strategy (SAS). How their design approach supports the Army's SAS architecture (see Attachment #3 Appendix A).

1.1.4. Tactical Assault Kit (TAK). How their design approach incorporates interoperability with TAK and Open Extended Reality XR.

1.2. Systems Engineering. Vendors should submit a narrative response which describes the following:

1.2.1. Iterative Development. Their approach to iterative development, to include, but not limited to how Soldier feedback will be incorporated into the development process and the use of virtual prototyping or similar techniques.

1.2.2. Modular Open Systems Approach (MOSA). How their MOSA will utilize an integrated business and technical strategy that employs a modular design and defines key interfaces using widely supported, consensus-based standards that are published and maintained by a recognized standards organization.

1.2.3. Model Based Systems Engineering (MBSE). Their approach to employing MBSE methodologies throughout the project lifecycle, including the creation, validation, and maintenance of system models as well as the software tools they will utilize for model creation (e.g., Visio, MagicDraw, etc.), and how these models will be used to support program and design reviews. Models should trace to system requirements, provide interface information, data flows, and utilize a black box approach to enable integration.

1.2.4. Design to Cost (DTC) Approach. Their approach to implementing DTC engineering to ensure affordability of the IVAS Next, to include the approach to minimizing both production unit costs and system lifecycle sustainment costs.

1.2.5. Software Development. Their software development process and describe how their process supports Continuous Integration/Continuous Delivery (CI/CD) in a Development, Security and Operations (DevSecOps) environment. Vendor should also describe how their software will be compatible with Android and Open XR.

1.2.6. Extensibility.

1.2.6.1. Software Development Kit (SDK). Their approach to developing an IVAS Next SDK functionality.

1.2.6.2. Application Developer Support. Their approach to providing technical support to Government and 3rd Party developers using the IVAS Next SDK to develop applications for the IVAS Next platform.

1.2.6.3. Application Validation & Verification (V&V). Their approach to providing V&V for Government and 3rd Party software applications developed using the IVAS Next SDK.

1.2.6.4. Virtual Training. Their proposed system's ability to host a mixed reality, immersive, virtual training capability via a government or 3rd party software application.

2.0. Program Management. Vendors should submit a narrative response which describes the following:

2.1. Program Management Approach. Their Program Management approach for their IVAS Next solution, to include but not limited to identifying teaming arrangements and key subcontractors and suppliers.

2.2 Security and Program Protection. Their approach to security and program protection, to include operational security, Supply Chain Risk Management (SCRM) and the handling of Controlled Unclassified Information (CUI) and classified information.

2.3. Cybersecurity. Their approach to cybersecurity.

2.4. Manufacturing Capability. Their capabilities and approach for sourcing parts and manufacturing the IVAS Next solution in prototype, production representative, follow-on production configurations, and an estimate on production capability for building the IVAS Next system.

2.5. Data Rights. Their data rights approach for their IVAS Next solution. Vendor should identify any Intellectual Property (technical data and/or software) they intend to assert at less than Government Purpose Rights (e.g., Limited Rights, Restricted Rights, etc.)

3.0 Quality Assurance, Reliability, and Test. Vendors should submit a narrative response which describes their proposed approach to Quality Assurance, reliability, and test for their IVAS Next solution.

4.0. Product Support. Vendors should submit a narrative response which describes their proposed product support approach for the IVAS Next to include, but not limited to, the following:

- Vendor's approach to supporting the Army's two-level maintenance concept.

- Vendor's approach to limiting the need for consumables (e.g., desiccant, batteries, etc.) and special tools.
- Vendor's approach to mitigating Diminishing Manufacturing Sources and Material Shortages (DMSMS).

5.0. **Program Risk.** Vendors should submit a narrative response which describes the following:

5.1. Risk Management and Initial Risk Assessment. Their approach to documenting, tracking, and managing areas of program risk. In addition, narrative should include a table with the vendor's initial assessment of program risks.

5.2. Relevant Experience. A summary of relevant experience within the last five (5) years. Relevant experience may include any of the following:

- Experience with developing or producing head borne augmented reality/mixed reality devices.
- Experience with developing or producing head borne night vision devices.
- Experience developing or producing other Soldier borne devices, to include weapon mounted sensors/sights or targeting devices; familiarization with weapon effects (e.g., ballistic shock).
- Experience developing or producing thermal or low light sensors.
- Experience developing or producing situational awareness software and/or systems.
- Experience developing mixed reality applications and experiences.
- Experience developing or producing virtual or augmented training devices and/or systems.

5.3. Performance.

5.3.1. Technology Readiness Level (TRL). The current TRL of their proposed solution and a narrative providing the basis for the TRL assessment.

5.3.2. Current Hardware Narrative. Their current hardware, to include identifying differences between their current hardware and their IVAS Next solution.

5.3.3. Current Hardware Performance. The current performance of their current hardware and Vendor should also **complete Attachment #3 Performance Matrix.**

5.3.4 Projected Performance. The projected performance of their IVAS Next solution at project completion. Vendor should also **complete Attachment #3 Performance Matrix.** Include any comments, question, or concerns regarding the Performance Matrix.

5.4. Challenge Areas. Vendor should provide a narrative describing any unique or creative solutions for the most complex and critical challenge areas relevant to an IVAS Next system.